UNIVERSITY OF AMAR TELIJI-LAGHOUAT

Skill exchange platform

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- **ABSTRACT:**

This project presents the design and implementation of a Skill Exchange Platform, a web-based system created to facilitate peer-to-peer knowledge sharing. In today's digital landscape, individuals often possess valuable skills but lack an organized environment to exchange them. Our platform addresses this gap by providing a collaborative ecosystem where users can register, showcase their expertise, and connect with others to trade skills in a mutual learning process.

The development of the platform followed a structured software engineering lifecycle. We conducted a comprehensive requirements analysis and designed a scalable database schema to manage user profiles, skill categories, and interactions. The system was implemented using modern web technologies, ensuring a responsive user interface (UI) and a robust back-end to handle authentication and data management (CRUD operations). The final result is a user-friendly platform that fosters community learning and professional development.

1 .INRODUCTION:

In the rapidly evolving digital era, knowledge has become the most valuable currency. Traditional learning methods are increasingly being supplemented by peer-to-peer interactions, where individuals exchange practical skills in a collaborative environment. However, finding the right person to trade skills with remains a challenge for many students and professionals. This project introduces the Skill Exchange Platform, a dedicated web-based solution designed to bridge the gap between skill seekers and skill providers. The platform allows users to

showcase their expertise, connect with others, and engage in a mutual learning process. By leveraging modern web technologies, our aim is to build a community-driven ecosystem that fosters continuous learning and professional growth. Throughout this report, we will detail the entire development lifecycle of the platform, starting from the initial requirements analysis and database design, moving to the architectural implementation of the back-end and front-end, and finally showcasing the user interface and system features.

2. Technologies Use:

Front-End: Vanilla JavaScript (ES6+), HTML5, and CSS3.
Justification: We opted for native technologies to ensure high performance, no external dependencies, and a deep understanding of DOM manipulation and the Fetch API.
Back-End: Node.js & Express.js.
Justification: Provides a scalable, non-blocking environment ideal for handling multiple API requests simultaneously.
Database: MongoDB.
Justification: A NoSQL approach allows for flexible data modeling, specifically for user profiles and skill listings.
Authentication: JSON Web Tokens (JWT) & Bcrypt.
Design Tools: CSS Grid, Flexbox, and Google Fonts (Tajawal).

3. System Design:

Architecture: The system follows a Client-Server architecture. The Front-End communicates with the Back-End via a RESTful API.

Database Schema:
User: Name, Email, Hashed Password.
Skill: Title, Description, Category, Owner (Reference to User ID).
API Endpoints:
POST /api/auth/register & login: User authentication.
GET /api/skills: Retrieve all available skills.
POST /api/skills: Create a new skill entry.
PUT/DELETE /api/skills/:id: Modify or remove skills (restricted to owners).

4.Implementation:

The application implements a full CRUD (Create, Read, Update, Delete) cycle for skill management. Key features include
State Management:
Using localStorage to persist JWT tokens, keeping users logged in across sessions.
Dynamic UI:
A custom-built routing system in JavaScript that switches between "Login", "Register", and "Dashboard" views without page reloads.
Security:
Middleware on the Back-End protects private routes, ensuring only authenticated users can post or edit content.
UX/UI: Responsive design ensures usability on mobile devices, complemented by auto-dismissing alerts and loading indicators.

5.GitHub Repository:

Link:

https://github.com/fabenadda-cell/projet-web

Structure:

/frontend: Contains HTML, CSS, and JS logic.

/backend: Contains Node.js server, routes, and controllers.

/models: Database schemas.

README.md: Documentation and setup instructions.

6. Challenges and Solutions：

Implementing a Single Page Application (SPA) behavior without frameworks like React.
Solution: Developed a custom JS function to toggle visibility of sections based on application state.
Challenge: Securely handling user passwords.
Solution: Integrated Bcrypt for one-way hashing before saving data to MongoDB.

7. Conclusion:

The Skill Exchange Platform successfully provides a functional and visually appealing solution for community learning. While the current version handles core features smoothly, future updates will focus on a real-time search engine, category filtering, and automated token expiration handling to further enhance security.

8. Contribution Table:

Member Name:	Contribution:
Benadda Fatima	Project Manager, Front-End (Main page, logic integration, team follow-up).
Bousbeaa Douha	Front-End Developer (Register & Login pages, Dashboard structure).
Zazaa Wafaa	CSS/UI Design (Responsive design, color palette, dark theme, styling).
Benmalek Ilyas	Back-End Development (API creation, Login system, CRUD logic).
Benniche Habiba	Database Design (Schemas, Table creation), Report Writing, README.

Note on Team Collaboration:

Although tasks were distributed among members, the final integration of the Front-End and Back-End was a collective

effort. The team worked together to connect the API endpoints, debug the system, and ensure that all components work seamlessly as a single integrated platform

Thanks for reading.